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Frontend 22-2
Batch

① What is Js and its features?

Javascript is a high level programming language used to make websites interactive and dynamic. Here interactive means the webpage can respond to user actions. Dynamic means the content of webpage can change automatically based on user actions or data.

Features of Js:

1. programming language - helps me add logic and functionality to a website.
2. Interpreted Language - When I write Js code and open my webpage in chrome, the browser's Javascript engine executes the code and shows the result. without manually compiling.
3. Dynamically typed
   ↓
   A variable can store diff types of values.
4. Dynamic Application - changing data without reloading the entire page
5. case-sensitive
6. Event-driven
7. Supports asynchronous programming
8. Api access - Js can communicate with servers to fetch and send data.
9. can also run in outside browsers using Node.Js

① What is Functions and types?

→ A Function is a reusable block of code used to perform a specific task. It runs whenever we call or invoke it. Once the function finishes its execution, its local variables usually cannot be accessed from outside the function.

→ A function can return a value using return statement. If no value is returned, JS returns undefined by default.

Syntax:

```
function functionName (parameters) {
    // code
    return value;
}
functionName(arguments);
```

Types

1. Function Declaration: It is a traditional way of defining a function using the function keyword a function name. It performs a specific task when the function is called.

2. Function Expression: When a function is stored inside a variable is called a function expression.

ex: let add = function() {
    // code

3. ~~Functi~~ Arrow function: It is a shorter way of writing a function in JS. It uses the arrow symbol => to define a function. It assigned to a variable using const or let and it is ES6 feature.

ex:
```
const add = (a,b) => {
    return a+b;
};
```

(5) What is spread operator ? When we use?

The spread operator uses 3 dots (...) to expand array elements or object properties into individual elements or properties. spread operator mostly used on array. and spread operator creates a shallow copy not a deep copy.

When do we use?

1. copying an array & objects
2. To combine array and objects
3. To add new elements to array and obj
4. Expand / unpack array and object.

(4) What is React and its features?

React JS is a modern Javascript library used to build interactive and dynamic user interfaces for web application. It was developed by Facebook (now meta). React helps us create reusable UI components

and webpage content efficiently.

Ex: In an online shopping website, React can update the cart when we add or remove products without reloading the entire page.

React have a 2 project setups — one is React → traditional React
Second is React VITE - Versions
↳ fast development build tool.

features:

1. component Based architecture - divide webpage into small reusable parts called components.
2. Virtual DOM - lightweight copy of actual DOM helps react update webpage changes efficiently.
3. Reusable components - Create once & reuse it in diff parts of application.
4. JSX - allows us to write HTML like code inside javascript
5. one way databinding - Data flows from parent to child through props.

(5) What is JSX ? Diff b/w JS and JSX.

→ Jsx stands for Javascript XML - It is ~~use~~ a syntax extension of Javascript that allows us to write HTML-like code inside Javascript. It makes creating and designing user interfaces easier in React. It also supports JS expressions inside curly braces { } and follows rules such as using className instead of class.

→ Jsx transformed into regular JS using tools like Babel.

Diff B/w Jsx and Js:

⇒ JS is a programming language used to add logic, functionality and interactivity to web Applications.

=> Jsx is a syntax extension of Javascript that allow us to write HTML-like code inside Javasoipt.

⑥ What is events? List out events name?

=> An event is an action performed by a user or detected by the browser. Examples include clicking a button, typing in an input field and moving the mouse over an element. Javascript can respond to these action using event listener Event helpers to create interactive and dynamic webpages.

=> common events are

Click → Button click
dbl click → Double click
mouseover → Mouse enters an element
mouse out → Mouse leaves an element
Input → Input value changes
Submit → Form submitted.

⑦ What is components & type of import and and write a program.

Solution:

→ A component in React is a small, reusable building block of a user interface. It helps us divide a webpage into smaller, manageable parts such as a header, navbar, sidebar and footer. login, Dashboard.

→ Each component can contain JSX, Javascript logic, event handlers, props and React hooks. Helps us to write maintainable and organized code.

→ Two types of component → Functional & class component

→ This is the main feature of react.

→ Import and export are used to share code btw Javascript files. Export is used to make variables, functions and classes available from one file and import is used to access them in another file.

→ Two main types: Named export and default export.

→ Named export allows us to export multiple times. and default export allows only one default export per file.

math.js

```
// Named export
export function add(a, b){
    return a+b;
}
// Default export
export default function greet() {
    return "Hello User";
}
```

app.js

Import

```
import greet, {add} from './math.js'
```